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United States Patent Application Publication

20250283677

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Cantrell; Christopher Thomas et al.

EXPENDABLE HEAT SINK SYSTEM AND CONTROL

Abstract

A cooling system includes a main coolant loop through which a first type of coolant (e.g., water or glycol) circulates to cool a heat load. During normal operation, cooling of the first type of coolant is performed by a heat exchanger using ambient or ram air. During peak loads, however, cooling of the first type of coolant is supplemental using an expendable coolant (e.g., supercritical carbon dioxide, liquid helium, liquid nitrogen, etc.). The expendable coolant can be vented from the cooling system after use in cooling the first type of coolant.

Inventors: Cantrell; Christopher Thomas (Jackson, MI), Wang; Hongbin (South Lyon, MI), Hemminger; Noah William (Ypsilanti, MI)

Applicant: Eaton Intelligent Power Limited (Dublin 4, IE)

Family ID: 1000008522802

Assignee: Eaton Intelligent Power Limited (Dublin 4, IE)

Appl. No.: 19/072163

Filed: March 06, 2025

Related U.S. Application Data

us-provisional-application US 63562358 20240307

Publication Classification

Int. Cl.: F28F27/02 (20060101); B64D33/10 (20060101)

U.S. Cl.:

CPC F28F27/02 (20130101); B64D33/10 (20130101);

Background/Summary

CROSS-REFERENCE PARAGRAPH [0001] This application claims the benefit of U.S. Provisional Application No. 63/562,358, filed Mar. 7, 2024, and titled “Expendable Heat Sink System and Control,” the disclosure of which is hereby incorporated herein by reference in its entirety.

BACKGROUND

[0002] Various components within an aircraft require thermal management to operate. For example, a propeller motor and a battery powering the propeller motor both generate heat during operation and can overheat (surpass an upper temperature limit) if left uncooled. Overheating may lead to electrical fires, equipment damage, performance, or useful life degradation. Certain components, such as the battery, also may have a lower temperature limit to operate properly. Accordingly, the aircraft is equipped with a thermal management system including cooling circuits for these components.

SUMMARY

[0003] Aspects of the disclosure are directed to a cooling system utilizing expendable media. In certain implementations, the cooling system can be used on an aircraft, such as a hybrid electric aircraft or electric aircraft. Other implementations are possible.

[0004] In certain aspects, expendable media is selectively released to cool a non-expendable coolant during peak heat loads. During non-peak heat loads, the non-expendable coolant is cooled using conventional means (e.g., ambient air). Because the expendable media acts as a thermal sink during peak heat loads, the non-expendable coolant and corresponding components need only be designed to accommodate normal operating loads, leading to smaller and lighter components.

[0005] In certain aspects, a first type of coolant is circulating along a main coolant loop to cool a heat load. The first type of coolant is cooled at an air-to-liquid heat exchanger. During select periods, the first coolant is diverted to a liquid-to-liquid heat exchanger at which the first type of coolant is cooled by a second type of coolant having a greater specific heat capacity. Because the second coolant cools only the first coolant and not the heat load directly, the components forming the heat load need not be designed to handle the pressure, temperature, or other characteristics of the second coolant.

[0006] In some implementations, the first type of coolant is directed back to the cooling circuit upstream of the air-to-liquid heat exchanger. In other implementations, the first type of coolant is directed back to the cooling circuit downstream of the air-to-liquid heat exchanger.

[0007] In some implementations, the second type of coolant is vented towards the air-to-liquid heat exchanger to further cool the first type of coolant. In other implementations, the second type of coolant is vented to ambient.

[0008] A variety of additional inventive aspects will be set forth in the description that follows. The inventive aspects can relate to individual features and to combinations of features. It is to be understood that both the forgoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the broad inventive concepts upon which the embodiments disclosed herein are based.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The accompanying drawings, which are incorporated in and constitute a part of the description, illustrate several aspects of the present disclosure. A brief description of the drawings is as follows:

[0010] FIG. 1 is a schematic diagram of a cooling system configured to be supplemented by an expendable media cooling circuit, the expendable media cooling circuit being integrated with the cooling system in a first configuration to supplement cooling of an air-to-liquid heat exchanger; [0011] FIG. 2 shows a flow path through the cooling system of FIG. 1 during normal operation; [0012] FIG. 3 shows a flow path through the cooling system and expendable media cooling circuit of FIG. 1 during peak heat loads; [0013] FIG. 4 shows an optional configuration in which the expendable media is vented towards the cooling system; [0014] FIG. 5 is a schematic diagram of a cooling system configured to be supplemented by an expendable media cooling circuit, the expendable media cooling circuit being integrated with the cooling system in a second configuration to replace the air-to-liquid heat exchanger; [0015] FIG. 6 shows a flow path through the cooling system of FIG. 5 during normal operation; and [0016] FIG. 7 shows a flow path through the cooling system and expendable media cooling circuit of FIG. 5 during peak heat loads.

DETAILED DESCRIPTION

[0017] Reference will now be made in detail to exemplary aspects of the present disclosure that are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

[0018] Referring to the figures in general, a cooling system **100** includes a main coolant loop **102** and a supplemental coolant path **120**. The main coolant loop **102** is configured to circulate a first coolant past a heat load **L**. The main coolant loop **102** includes a coolant pump **104** and a first heat exchanger **106** (e.g., an air-to-liquid heat exchanger) at which heat is removed from the first coolant. For example, ambient air or ram air **108** may be blown over the first exchanger **106** to cool the first coolant.

[0019] The supplemental coolant path **120** includes a tank **122** and a second heat exchanger **126**. In certain implementations, the second heat exchanger **126** is a liquid-to-liquid heat exchanger. As the term is used herein, a liquid-to-liquid heat exchanger cools a first liquid coolant using a second coolant that is either liquid or in a supercritical state. The tank **122** holds a second coolant **EC** having a specific heat capacity that is greater than the specific heat capacity of the first coolant. In other words, the second coolant is capable of absorbing a greater amount of heat than the first coolant for a given time period. The supplemental coolant path **120** includes a valve arrangement **124** to selectively release the second coolant **EC** from the tank **122**. In some implementations, the valve arrangement **124** is located at the tank **122**. In other implementations, the valve arrangement **124** is downstream of the tank **122**.

[0020] In some implementations, the second coolant **EC** is held in a liquid state within the tank **122**. In certain implementations, a valve arrangement **124** and/or a valve arrangement **128** operates to release the second coolant **EC** from the tank **122** and to ensure a desired phase of the second coolant **EC**. For example, the valve arrangement **128** controls when the second coolant **EC** is vented from the supplemental coolant path **120**. By remaining closed, the pressure within the supplemental coolant path **120** increases, which may bring the second coolant **EC** to a supercritical state.

[0021] The supplemental coolant path directs the released second coolant through the second heat exchanger **126**. In certain implementations, the second coolant **EC** is vented out of the cooling system **100** after passing through the second heat exchanger **126**. For example, the valve arrangement **128** may intermittently vent the second coolant **EC**. In some examples, the second coolant **EC** is vented to ambient (e.g., see FIG. 3). In other examples, the second coolant **EC** is vented towards the first heat exchanger **106** to act as ram air (e.g., see FIG. 4).

[0022] A valve arrangement **110**, **110'** is configured to selectively route the first coolant of the main coolant loop **102** to the second heat exchanger **126** of the supplemental coolant path **120** to be

cooled by the second coolant EC. The valve arrangement **110**, **110'** includes a first valve **111** that selectively directs the first coolant either towards the second heat exchanger **126** or towards the first heat exchanger **106**. In certain implementations, the valve arrangement **110**, **110'** also includes a second valve **112**, **112'** that receives the first coolant after the first coolant has been cooled at the second heat exchanger **126**. The second valve **112**, **112'** integrates the cooled first coolant back into the main coolant loop **102**.

[0023] In some implementations, the second valve **112** is located upstream of the first heat exchanger **106** (e.g., see FIG. 1). In such implementations, the cooled first coolant is routed through the first heat exchanger **106** for further cooling (e.g., see FIG. 3). In other implementations, the second valve **112'** is located downstream of the first heat exchanger **106** (e.g., see FIG. 5). In such implementations, the cooled first coolant bypasses the first heat exchanger **106** before heading to the heat load L (e.g., see FIG. 7). In still other implementations, the first coolant is directed back into the main coolant loop **102** without a separate valve.

[0024] Suitable examples of a first type of coolant include water, ethylene glycol, or propylene glycol. Suitable examples of a second type of coolant EC include supercritical carbon dioxide (sCO.sub.2), liquid hydrogen (LH.sub.2), and liquid nitrogen (LN.sub.2). In certain implementations, the second coolant has a specific heat capacity that is substantially greater than the specific heat capacity of the first coolant. For example, the second coolant may have a specific heat capacity that is 2 times, 5 times, 10 times, 20 times, or 30 times the specific heat capacity of the first coolant at a particular temperature.

[0025] In certain implementations, the second coolant is maintained at a higher pressure than the first coolant. Accordingly, components of the main coolant loop **102** need not be built to withstand the pressure levels of the second coolant.

[0026] In certain examples, the first heat exchanger **106** has a plate/fin (PFHE) construction. In certain examples, the second heat exchanger **126** includes a stacked plate heat exchanger.

[0027] In certain implementations, a controller **140** manages operation of the valve arrangement **110**. In certain examples, the controller **140** manages operation of the first valve **111** to selectively direct the first coolant towards the second heat exchanger **126**. In certain examples, the controller **140** manages operation of the second valve **112**, **112'** that selectively releases the second coolant EC from the tank **122**. In certain examples, the controller **140** manages operation of the valve arrangement **124** that selectively releases the second coolant EC from the tank **122**.

[0028] In certain implementations, the controller **140** manages the operation of the valve arrangements **110**, **124** based on one or more characteristics of the first and second coolants. In certain implementations, one or more sensors **114** (e.g., temperature sensor, pressure sensor, etc.) are disposed throughout the main coolant loop **102** and/or the supplemental coolant path **120**. In certain examples, the controller **140** will open the valve **124** of the supplemental coolant path **120** to release the second coolant EC from the tank **122** towards the second heat exchanger **126**. In certain examples, the controller **140** may configure the valve arrangement **110** to direct the first coolant towards the second heat exchanger **126** based on a temperature reading of the first coolant taken by a temperature sensor **114** along the main coolant path **102**.

[0029] In some examples, the controller **140** may operate the first valve **111** to direct the first coolant towards the second heat exchanger **126** when the temperature of the first coolant exceeds a predetermined level. For example, the first coolant may be directed towards the second heat exchanger **126** when the temperature of the first coolant reaches 35 degrees Celsius, 40 degrees Celsius, 45 degrees Celsius, 50 degrees Celsius, 60 degrees Celsius, 70 degrees Celsius, 80 degrees Celsius, or 85 degrees Celsius. In other examples, the controller **140** may operate the first valve **111** to direct the coolant towards the second heat exchanger **126** when the heat load L reaches a predetermined level (e.g., 35, 40, 45, 50, 60, 70, 80, or 85 degrees Celsius).

[0030] In certain examples, the controller **140** may configure the valve arrangement **128** to direct the second coolant EC leaving the second heat exchanger **126** towards the first heat exchanger **106**

when the temperature of the second coolant EC remain below a predetermined threshold and/or when the temperature of the first coolant rises above a predetermined threshold. For example, a temperature sensor **114** may be disposed downstream of the second heat exchanger **126** along the supplemental coolant path **120**. If the second coolant EC leaving the second heat exchanger **126** is sufficiently cold, then the second coolant EC may be vented at the first heat exchanger **106** as ram air to further cool the first coolant. In certain examples, the second coolant EC may be vented to the first heat exchanger **106** if suitable ram air is not available (e.g., if ambient air temperatures are too high). Alternatively, if the temperature of the second coolant EC leaving the second heat exchanger **126** is too high and/or if the temperature of the first coolant is below a predetermined threshold, then the controller **140** may configure the valve arrangement **128** to vent the second coolant EC to ambient.

[0031] Example implementations of the disclosure are further described in the below aspects.

[0032] Aspect 1. A cooling system comprising: [0033] a main coolant loop configured to circulate a first coolant having a first specific heat capacity, the main coolant loop including a coolant pump and a first heat exchanger, the main coolant loop extending past a heat load; [0034] a supplemental coolant path including a tank and a second heat exchanger, the tank holding a second coolant having a second specific heat capacity that is greater than the first specific heat capacity of the first coolant; and [0035] a valve arrangement configured to selectively route the first coolant of the main coolant loop to the second heat exchanger of the supplemental coolant path to be cooled by the second coolant.

[0036] Aspect 2. The cooling system of aspect 1, wherein the valve arrangement is configured to route the first coolant back to the main coolant loop upstream of the first heat exchanger after being cooled by the second heat exchanger.

[0037] Aspect 3. The cooling system of aspect 1, wherein the valve arrangement is configured to route the first coolant back to the main coolant loop downstream of the first heat exchanger after being cooled by the second heat exchanger.

[0038] Aspect 4. The cooling system of any of aspects 1-3, wherein the second coolant is vented after flowing through the second heat exchanger.

[0039] Aspect 5. The cooling system of aspect 4, wherein the vented second coolant is directed toward the first heat exchanger to cool the first coolant.

[0040] Aspect 6. The cooling system of any of aspects 1-5, further comprising a controller configured to operate the valve arrangement.

[0041] Aspect 7. The cooling system of aspect 6, wherein the controller operates the valve arrangement based on a temperature of the first coolant.

[0042] Aspect 8. The cooling system of aspect 6, wherein the controller operates a valve arrangement of the supplemental coolant path to selectively release the second coolant to the second heat exchanger.

[0043] Aspect 9. The cooling system of aspect 6, wherein the controller operates a valve arrangement of the supplemental coolant path to selectively vent the second coolant.

[0044] Aspect 10. The cooling system of any of aspects 1-9, wherein the first heat exchanger is an air-to-liquid heat exchanger and wherein the second heat exchanger is a liquid-to-liquid heat exchanger.

[0045] Aspect 11. A method of cooling a heat load, the method comprising: [0046] circulating a first coolant between the heat load and a first heat exchanger along a main coolant loop; [0047] determining that the first heat exchanger is not sufficiently cooling the first coolant; [0048] redirecting the first coolant to a second heat exchanger; [0049] directing a second coolant to the second heat exchanger, the second coolant being colder than the first coolant; and [0050] directing the first coolant back to the main coolant loop for further circulation.

[0051] Aspect 12. The method of aspect 11, further comprising venting the second coolant.

[0052] Aspect 13. The method of aspect 12, wherein venting the second coolant comprises venting

the second coolant to ambient.

[0053] Aspect 14. The method of aspect 12, wherein venting the second coolant comprises venting the second coolant towards the first heat exchanger.

[0054] Aspect 15. The method of aspect 11, wherein the second coolant has a greater specific heat capacity than the first coolant.

[0055] Aspect 16. The method of aspect 11, wherein determining that the first heat exchanger is not sufficiently cooling the first coolant comprises analyzing a temperature reading of the first coolant obtained along the main coolant loop.

[0056] Aspect 17. A cooling system for a heat load, the cooling system comprising: [0057] a main coolant loop configured to circulate a non-expendable coolant, the main coolant loop including a coolant pump and a first heat exchanger, the main coolant loop circulating the non-expendable coolant past the heat load; [0058] a supplemental coolant path including a tank holding an expendable coolant, the supplemental coolant path also including a second heat exchanger through which the expendable coolant is selectively directed; and [0059] a controller configured to manage releasing of the expendable coolant to the second heat exchanger and configured to manage redirecting the non-expendable coolant to the second heat exchanger.

[0060] Aspect 18. The cooling system of aspect 17, further comprising a plurality of sensors disposed along the main coolant loop and the supplemental coolant path.

[0061] Aspect 19. The cooling system of aspect 17, further comprising a first valve arrangement configured to manage a selective connection between the main coolant loop and the supplemental coolant path, wherein the controller manages operation of the first valve arrangement.

[0062] Aspect 20. The cooling system of any of aspects 17-19, further comprising a second valve arrangement configured to manage selective venting of the expendable coolant, wherein the controller manages operation of the second valve arrangement.

[0063] Having described the preferred aspects and implementations of the present disclosure, modifications and equivalents of the disclosed concepts may readily occur to one skilled in the art. However, it is intended that such modifications and equivalents be included within the scope of the claims which are appended hereto.

Claims

1. A cooling system comprising: a main coolant loop configured to circulate a first coolant having a first specific heat capacity, the main coolant loop including a coolant pump and a first heat exchanger, the main coolant loop extending past a heat load; a supplemental coolant path including a tank and a second heat exchanger, the tank holding a second coolant having a second specific heat capacity that is greater than the first specific heat capacity of the first coolant; and a valve arrangement configured to selectively route the first coolant of the main coolant loop to the second heat exchanger of the supplemental coolant path to be cooled by the second coolant.

2. The cooling system of claim 1, wherein the valve arrangement is configured to route the first coolant back to the main coolant loop upstream of the first heat exchanger after being cooled by the second heat exchanger.

3. The cooling system of claim 1, wherein the valve arrangement is configured to route the first coolant back to the main coolant loop downstream of the first heat exchanger after being cooled by the second heat exchanger.

4. The cooling system of claim 1, wherein the second coolant is vented after flowing through the second heat exchanger.

5. The cooling system of claim 4, wherein the vented second coolant is directed toward the first heat exchanger to cool the first coolant.

6. The cooling system of claim 1, further comprising a controller configured to operate the valve arrangement.

7. The cooling system of claim 6, wherein the controller operates the valve arrangement based on a temperature of the first coolant.
 8. The cooling system of claim 6, wherein the controller operates a valve arrangement of the supplemental coolant path to selectively release the second coolant to the second heat exchanger.
 9. The cooling system of claim 6, wherein the controller operates a valve arrangement of the supplemental coolant path to selectively vent the second coolant.
 10. The cooling system of claim 1, wherein the first heat exchanger is an air-to-liquid heat exchanger and wherein the second heat exchanger is a liquid-to-liquid heat exchanger.
 11. A method of cooling a heat load, the method comprising: circulating a first coolant between the heat load and a first heat exchanger along a main coolant loop; determining that the first heat exchanger is not sufficiently cooling the first coolant; redirecting the first coolant to a second heat exchanger; directing a second coolant to the second heat exchanger, the second coolant being colder than the first coolant; and directing the first coolant back to the main coolant loop for further circulation.
 12. The method of claim 11, further comprising venting the second coolant.
 13. The method of claim 12, wherein venting the second coolant comprises venting the second coolant to ambient.
 14. The method of claim 12, wherein venting the second coolant comprises venting the second coolant towards the first heat exchanger.
 15. The method of claim 11, wherein the second coolant has a greater specific heat capacity than the first coolant.
 16. The method of claim 11, wherein determining that the first heat exchanger is not sufficiently cooling the first coolant comprises analyzing a temperature reading of the first coolant obtained along the main coolant loop.
 17. A cooling system for a heat load, the cooling system comprising: a main coolant loop configured to circulate a non-expendable coolant, the main coolant loop including a coolant pump and a first heat exchanger, the main coolant loop circulating the non-expendable coolant past the heat load; a supplemental coolant path including a tank holding an expendable coolant, the supplemental coolant path also including a second heat exchanger through which the expendable coolant is selectively directed; and a controller configured to manage releasing of the expendable coolant to the second heat exchanger and configured to manage redirecting the non-expendable coolant to the second heat exchanger.
 18. The cooling system of claim 17, further comprising a plurality of sensors disposed along the main coolant loop and the supplemental coolant path.
 19. The cooling system of claim 17, further comprising a first valve arrangement configured to manage a selective connection between the main coolant loop and the supplemental coolant path, wherein the controller manages operation of the first valve arrangement.
 20. The cooling system of claim 17, further comprising a second valve arrangement configured to manage selective venting of the expendable coolant, wherein the controller manages operation of the second valve arrangement.
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